

AP CALCULUS AB - UNIT 5

Guided Notes 5.10: Optimization Problems



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we identify an objective quantity and a constraint relating variables?

- Identify an objective quantity and a constraint relating variables.
- Rewrite the objective as a one-variable function on a meaningful domain.
- Find and justify an absolute maximum or minimum using critical points and endpoints.
- Interpret optimized dimensions and values with correct units.

Key Knowledge, Vocabulary, and Notation

Objective

The quantity to maximize or minimize, such as area, volume, cost, distance, or time.

Constraint

The equation linking the variables and fixed data.

Feasible domain

Input values allowed by geometry, physics, and the context.

Candidate set

Interior critical points and relevant endpoints or limiting boundary cases.

Verification

Use sign analysis, second derivative, or comparison of objective values to prove optimality.

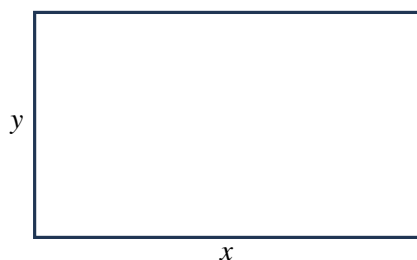
Interpretation

Report dimensions and the optimized quantity, including units and appropriate rounding.

Concept Development and Strategy

1. Draw and label a model, then state the objective and constraint separately.
2. Use the constraint to eliminate variables until the objective depends on one variable.
3. Write the physically meaningful domain before differentiating.
4. Find critical points and compare all feasible candidates; do not stop after solving $F' = 0$.
5. Answer the original question, not merely the value of the chosen variable.

Shared Representation / Data



Objective: $A = xy$
 Constraint: $2x + 2y = P$
 Domain: $0 < x < P/2$

Worked Example: Maximum rectangle

A rectangle has perimeter 40 m. Find dimensions of maximum area.

Answer and Reasoning

$2x + 2y = 40$, so $y = 20 - x$. $A = x(20 - x)$ on $0 < x < 20$. $A' = 20 - 2x = 0$ gives $x = 10, y = 10$; the maximum area is 100 m^2 .

Worked Example: Open-top box

Squares of side x are cut from a 20 cm by 14 cm sheet. Maximize the box volume.

Answer and Reasoning

$V = x(20 - 2x)(14 - 2x)$ on $0 < x < 7$. $V' = 12x^2 - 136x + 280 = 4(3x^2 - 34x + 70)$. The feasible critical point is $x = (17 - \sqrt{79})/3 \approx 2.704 \text{ cm}$; compare endpoints to verify the maximum.

Worked Example: Minimum material cylinder

A closed cylinder has volume $500\pi \text{ cm}^3$. Express surface area in one variable.

Answer and Reasoning

$\pi r^2 h = 500\pi$, so $h = 500/r^2$. Then $S(r) = 2\pi r^2 + 2\pi r(500/r^2) = 2\pi r^2 + 1000\pi/r, r > 0$.

Worked Example: Closest point

Find the point on $y = x^2$ closest to $(0, 3)$.

Answer and Reasoning

Minimize squared distance $D^2 = x^2 + (x^2 - 3)^2$. Derivative $2x + 4x(x^2 - 3) = 2x(2x^2 - 5)$. Candidates $x = 0, \pm\sqrt{5/2}$; comparison gives closest points $(\pm\sqrt{5/2}, 5/2)$.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. For a rectangle with fixed perimeter P , express y in terms of x .

2. State the feasible domain for rectangle width x in the previous problem.

3. A rectangle has area 96 m^2 . Express perimeter as a function of width x .

4. Find dimensions minimizing that perimeter.

5. Find the maximum area in the river-fencing problem.

6. Optimize the poster dimensions from the previous problem.

7. A rectangle is inscribed under $y = 12 - x^2$ above the x -axis, symmetric about the y -axis. Maximize area.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. The objective function is

- | | |
|-----------------------------|--|
| (A) the fixed equation only | (B) the quantity to maximize or minimize |
| (C) the derivative | (D) the domain |

2. After finding $F'(x) = 0$, the next essential step is

- | | |
|--|--|
| (A) stop | (B) ignore endpoints |
| (C) verify and compare feasible candidates | (D) differentiate the constraint again |

3. A rectangle has area 150 and minimum perimeter. Each side is closest to

- | | |
|-----------|-----------|
| (A) 10.00 | (B) 12.25 |
| (C) 15.00 | (D) 18.37 |

Common Errors and AP Precision

- A zero derivative or zero second derivative is a candidate condition, not by itself a complete extremum or inflection justification.
- State theorem hypotheses, include endpoints in absolute-extrema work, and verify every optimization candidate on the feasible domain.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

Prove among rectangles with fixed area, the square has minimum diagonal.

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